Changing Variables in Double Integrals: Geometry, Jacobians, and Multiplicity

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The bounds of an iterated integral are descriptions of a region, not decorations attached to the integration symbols. A good coordinate change transforms a difficult region into a simple one, but it must also measure how area changes. The determinant of the derivative gives this local area factor. When the coordinate map covers a point several times, an additional global issue appears: multiplicity.[1]

§1.1 A reversed integral should describe the same region

Consider the region between $y = x^2$ and $y = x$ for $0 \le x \le 1$. It is

$$D = \{(x, y) : 0 \le x \le 1,\ x^2 \le y \le x\}.$$

For each fixed x, the vertical slice has length $x - x^2$, so

$$\operatorname{area}(D) = \int_0^1 (x - x^2)\, dx = \frac{1}{6}.$$

If we choose horizontal slices, the conditions $x^2 \le y \le x$ become

$$y \le x \le \sqrt{y}, \qquad 0 \le y \le 1.$$

Therefore the same area is

$$\boxed{\int_0^1 \int_{x^2}^{x} 1\, dy\, dx = \int_0^1 \int_{y}^{\sqrt{y}} 1\, dx\, dy = \frac{1}{6}.}$$

Nothing about the integrand changed. We merely described the same measurable set as a union of different one-dimensional fibers. The equality is an instance of Fubini's theorem.

§1.2 Jacobian as an Area Scale

Let $\Phi(u, v) = (x(u, v), y(u, v))$ be a continuously differentiable map. Near a point (u_0, v_0), a tiny rectangle of sides du, dv is sent, to first order, to a parallelogram spanned by

$$\Phi_u\, du, \qquad \Phi_v\, dv.$$

Its area is

$$\left| \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix} \right| du\, dv.$$

[1] For Fubini and the multivariable change-of-variables theorem, see MIT 18.02SC, *Multivariable Calculus*, Units 4–5: https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/. The general multiplicity formula is discussed in Evans and Gariepy, *Measure Theory and Fine Properties of Functions*.

Thus the local area scale is the absolute Jacobian determinant

$$J_\Phi = \det D\Phi.$$

For a one-to-one C^1 coordinate change with nonzero Jacobian on a suitable region, the change-of-variables theorem reads

$$\boxed{\iint_{\Phi(U)} f(x,y)\,dx\,dy = \iint_U f(\Phi(u,v))|J_\Phi(u,v)|\,du\,dv.}$$

The absolute value ensures that ordinary area is positive whether the coordinate transformation preserves or reverses orientation. For oriented differential forms, the sign of the determinant is retained instead.

§1.3 Coordinates adapted to two families of straight lines

Suppose a region is bounded by lines of the form $x+y=$ constant and $x-y=$ constant. Set

$$u = x+y, \qquad v = x-y.$$

Then

$$x = \frac{u+v}{2}, \qquad y = \frac{u-v}{2}, \qquad \det\frac{\partial(x,y)}{\partial(u,v)} = -\frac{1}{2}.$$

Consider the parallelogram

$$D = \{(x,y) : 0 \le x+y \le 2, \quad 0 \le x-y \le 1\}.$$

It becomes the rectangle $0 \le u \le 2,\ 0 \le v \le 1$. Its area is immediately

$$\operatorname{area}(D) = \int_0^2 \int_0^1 \frac{1}{2}\,dv\,du = 1.$$

The same substitution simplifies nonconstant integrands. Since

$$x^2 - y^2 = (x+y)(x-y) = uv,$$

we get

$$\boxed{\iint_D (x^2-y^2)\,dx\,dy = \frac{1}{2}\int_0^2 \int_0^1 uv\,dv\,du = \frac{1}{2}.}$$

The region and the integrand become simple at the same time. This is the real criterion for a useful substitution.

§1.4 Polar coordinates derive a Gaussian integral

The polar map is

$$\Phi(r,\theta) = (r\cos\theta, r\sin\theta), \qquad r \ge 0, \quad 0 \le \theta < 2\pi.$$

Its derivative has determinant

$$\det\begin{pmatrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{pmatrix} = r.$$

Thus, away from the origin and the seam at $\theta = 0$, polar coordinates give

$$dx\,dy = r\,dr\,d\theta.$$

The factor r is not arbitrary: angular displacements of size $d\theta$ sweep arcs whose length is approximately $r\,d\theta$.

A particularly elegant use is to evaluate

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx.$$

The integral is positive and finite. Tonelli's theorem gives

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy.$$

Polar coordinates convert the radial Gaussian into a one-dimensional exponential:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta = 2\pi\cdot\frac{1}{2} = \pi.$$

Hence

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.}$$

The two-dimensional detour solves a one-dimensional integral because the product of two Gaussian factors becomes rotationally symmetric.

§1.5 Global Overcounting by a Jacobian

The usual formula needs a one-to-one parametrization, at least apart from negligible boundary sets. To see why, identify the plane with the complex numbers and consider

$$\Phi(z) = z^2.$$

On the annulus $A = \{z : 1 < |z| < 2\}$, the map is smooth and its real Jacobian is

$$|J_\Phi(z)| = |\Phi'(z)|^2 = 4|z|^2.$$

Its image is the annulus $B = \{w : 1 < |w| < 4\}$. But each point of B has *two* preimages in A, the two square roots.

If we blindly apply the one-to-one formula to the constant function 1, we obtain

$$\int_A 4|z|^2\,dA = \int_0^{2\pi}\int_1^2 4r^3\,dr\,d\theta = 30\pi.$$

The actual area of B is

$$|B| = \pi(4^2 - 1^2) = 15\pi.$$

The integral with the Jacobian is twice the image area, exactly because the annulus is covered twice:

$$\boxed{\int_A |J_\Phi(z)|\,dA = 2\,|\Phi(A)|.}$$

More generally, the area formula for a smooth finite-to-one map involves the number of preimages of almost every image point. A nonzero local Jacobian guarantees local invertibility, but it does not guarantee global one-to-one behavior.

§1.6 Coarea counts fibers of a nonlinear function

The same distinction appears when the new variable is a scalar function rather than a full coordinate system. For a sufficiently regular $g : \mathbb{R}^2 \to \mathbb{R}$, the coarea formula relates a planar integral to integrals over the level curves $g^{-1}(t)$:

$$\int_{\mathbb{R}^2} h(x)|\nabla g(x)|\,dx = \int_{\mathbb{R}} \left(\int_{g^{-1}(t)} h(x)\,d\mathcal{H}^1(x) \right) dt.$$

For $g(x, y) = x^2 + y^2$, the level $g = t > 0$ is a circle of radius $\sqrt{t}$, and $|\nabla g| = 2\sqrt{t}$. Consequently a radial function $\psi(g)$ has

$$\begin{aligned} \int_{\mathbb{R}^2} \psi(x^2 + y^2)\,dx\,dy &= \int_0^\infty \psi(t) \frac{2\pi\sqrt{t}}{2\sqrt{t}}\,dt \\ &= \boxed{\pi \int_0^\infty \psi(t)\,dt}. \end{aligned}$$

Taking $\psi(t) = e^{-t}$ recovers the Gaussian calculation.

The lesson extends from elementary region sketches to geometric measure theory. Reversing integration order changes the slicing map; the Jacobian measures local area distortion; the area formula accounts for how often a parametrization covers the image; and coarea integrates over the fibers of a nonlinear function. Before making a substitution, the decisive questions are what the new fibers look like, how they scale, and whether they overlap.